

Knapsack problem

Knapsack

Given a set of items, each with a weight and a value, determine the which items to include in a collection so that the total weight is less than or equal to a given limit and the total value is as large as possible.

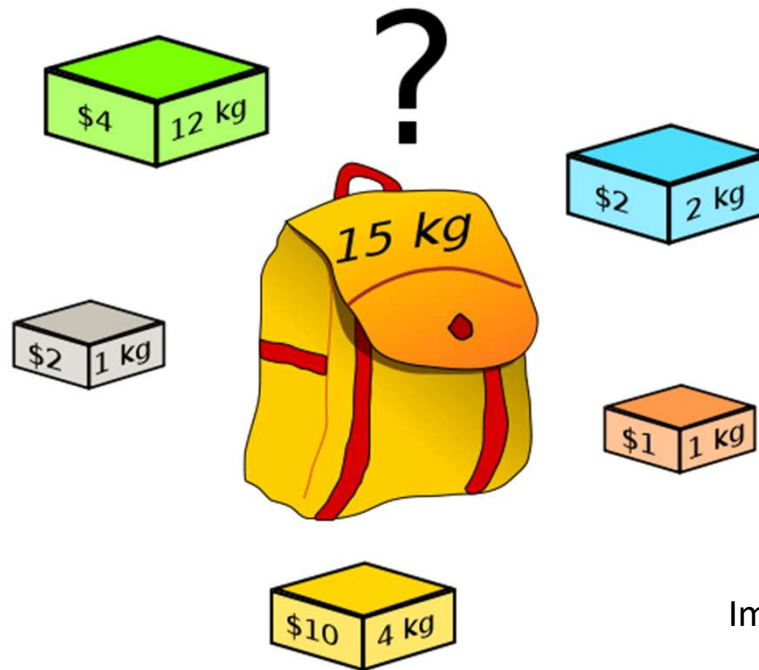


Image Source: Wikipedia

What is an optimal solution to this problem?

Problem 16.4 (*S*) Consider an instance of the knapsack problem with five items:

Item	Value	Size
1	1	1
2	2	3
3	3	2
4	4	5
5	5	4

and knapsack capacity $C = 9$. What are the final array entries of the Knapsack algorithm from Section 16.5, and which items belong to the optimal solution?

Knapsack problem

Inputs:

- n is number of items
 - value of i th item is v_i
 - size of i th item is s_i
 - Knapsack capacity is C
-
- Output: a subset of items

Knapsack problem

Inputs:

- n is number of items
 - value of i th item is v_i
 - size of i th item is s_i
 - Knapsack capacity is C
-
- Output: a subset of items

Goal: Maximize total value of the items
while still fitting in the knapsack,
total size of items should be $\leq C$.

Problem definition

Problem: Knapsack

Input: Item values $v_1, v_2, \dots, v_n$, item sizes $s_1, s_2, \dots, s_n$, and a knapsack capacity C . (All positive integers.)

Output: A subset $S \subseteq \{1, 2, \dots, n\}$ of items with the maximum-possible sum $\sum_{i \in S} v_i$ of values, subject to having total size $\sum_{i \in S} s_i$ at most C .

Quiz 16.5

Consider an instance of the knapsack problem with knapsack capacity $C = 6$ and four items:

Item	Value	Size
1	3	4
2	2	3
3	4	2
4	4	3

What is the total value of an optimal solution?

- a) 6
- b) 7
- c) 8
- d) 10

Solution to Quiz 16.5

Correct answer: (c). Because the knapsack capacity is 6, there is no room to choose more than two items. The most valuable pair of items is the third and fourth ones (with total value 8), and these fit in the knapsack (with total size 5).

Optimal substructure in Knapsack solution

Consider an instance of the knapsack problem with item values $v_1, v_2, \dots, v_n$, item sizes $s_1, s_2, \dots, s_n$, and knapsack capacity C , and suppose someone handed us on a silver platter an optimal solution $S \subseteq \{1, 2, \dots, n\}$ with total value $V = \sum_{i \in S} v_i$. What must it look like? As with the WIS problem, we start with a tautology: S either contains the last item (item n) or it doesn't.¹⁸

Last item not in the desired subset

- Given a solution with total size $\leq C$

[illegible]

Case 1: $n \notin S$. Because the optimal solution S excludes the last item, it can be regarded as a feasible solution (still with total value V and total size at most C) to the smaller problem consisting of only the first $n - 1$ items (and knapsack capacity C). Moreover, S must be an optimal solution to the smaller subproblem: If there were a solution $S^* \subseteq \{1, 2, \dots, n - 1\}$ with total size at most C and total value greater than V , it would also constitute such a solution in the original instance. This would contradict the supposed optimality of S .

Case 2

- Last item is in the solution

Case 2: $n \in S$. The trickier case is when the optimal solution S makes use of the last item n . This case can occur only when $s_n \leq C$. We can't regard S as a feasible solution to a smaller problem with only the first $n - 1$ items, but we can after removing item n . Is $S - \{n\}$ an optimal solution to a smaller subproblem?

Quiz 16.6

Which of the following statements hold for the set $S - \{n\}$?
(Choose all that apply.)

- a) It is an optimal solution to the subproblem consisting of the first $n - 1$ items and knapsack capacity C .
- b) It is an optimal solution to the subproblem consisting of the first $n - 1$ items and knapsack capacity $C - v_n$.
- c) It is an optimal solution to the subproblem consisting of the first $n - 1$ items and knapsack capacity $C - s_n$.
- d) It might not be feasible if the knapsack capacity is only $C - s_n$.

Which of the following statements hold for the set $S - \{n\}$?
(Choose all that apply.)

- a) It is an optimal solution to the subproblem consisting of the first $n - 1$ items and knapsack capacity C .

²⁵For example, suppose $C = 2$ and consider two items, with $v_1 = s_1 = 1$ and $v_2 = s_2 = 2$. The optimal solution S is $\{2\}$. $S - \{2\}$ is the empty set, but the only optimal solution to the subproblem consisting of the first item and knapsack capacity 2 is $\{1\}$.

Solution to Quiz 16.6

Correct answer: (c). The most obviously false statement is (b), which doesn't even typecheck (C is in units of size, v_n in units of value). For example, v_n could be bigger than C , in which case $C - v_n$ is negative and meaningless. For (d), because S is feasible for the original problem, its total size is at most C ; after n is removed from S , the total size drops to at most $C - s_n$ and, hence, $S - \{n\}$ is feasible for the reduced capacity. Answer (a) is a natural guess but is also incorrect.²⁵

In (c), we are effectively reserving s_n units of capacity for item n 's inclusion, which leaves a residual capacity of $C - s_n$. $S - \{n\}$ is a feasible solution to the smaller subproblem (with knapsack capacity $C - s_n$) with total value $V - v_n$. If there were a better solution $S^* \subseteq \{1, 2, \dots, n - 1\}$, with total value $V^* > V - v_n$ and total size at most $C - s_n$, then $S^* \cup \{n\}$ would have total size at most C and total value $V^* + v_n > (V - v_n) + v_n = V$. This would contradict the supposed optimality of S for the original problem.

Lemma 16.4 (Knapsack Optimal Substructure) *Let S be an optimal solution to a knapsack problem with $n \geq 1$ items, item values $v_1, v_2, \dots, v_n$, item sizes $s_1, s_2, \dots, s_n$, and knapsack capacity C . Then, S is either:*

- (i) an optimal solution for the first $n - 1$ items with knapsack capacity C ; or*
- (ii) an optimal solution for the first $n - 1$ items with knapsack capacity $C - s_n$, supplemented with the last item n .*

Value Table to store intermediate results

[illegible][illegible]

Corollary 16.5 (Knapsack Recurrence) *With the assumptions and notation of Lemma 16.4, let $V_{i,c}$ denote the maximum total value of a subset of the first i items with total size at most c . (When $i = 0$, interpret $V_{i,c}$ as 0.) For every $i = 1, 2, \dots, n$ and $c = 0, 1, 2, \dots, C$,*

$$V_{i,c} = \begin{cases} \underbrace{V_{i-1,c}}_{\text{Case 1}} & \text{if } s_i > c \\ \max\{\underbrace{V_{i-1,c}}_{\text{Case 1}}, \underbrace{V_{i-1,c-s_i} + v_i}_{\text{Case 2}}\} & \text{if } s_i \leq c. \end{cases}$$

Because both c and items' sizes are integers, the residual capacity $c - s_i$ in the second case is also an integer.

Subproblems parameterized by two indices:

- length i of the prefix of available items and
- the available knapsack capacity c .

Knapsack: Subproblems

Compute $V_{i,c}$, the total value of an optimal knapsack solution with the first i items and knapsack capacity c .

(For each $i = 0, 1, 2, \dots, n$ and $c = 0, 1, 2, \dots, C$.)

²⁰In the WIS problem on path graphs, there's only one dimension in which a subproblem can get smaller (by having fewer vertices). In the knapsack problem, there are two (by having fewer items, or less knapsack capacity).

Input: item values $v_1, \dots, v_n$, item sizes $s_1, \dots, s_n$, and a knapsack capacity C (all positive integers).
Output: the maximum total value of a subset $S \subseteq \{1, 2, \dots, n\}$ with $\sum_{i \in S} s_i \leq C$.

```
// subproblem solutions (indexed from 0)
A := (n + 1) × (C + 1) two-dimensional array
// base case (i = 0)
for c = 0 to C do
    A[0][c] = 0
// systematically solve all subproblems
for i = 1 to n do
    for c = 0 to C do
        // use recurrence from Corollary 16.5
        if s_i > c then
            A[i][c] := A[i - 1][c]
        else
            A[i][c] :=
                max{  $\underbrace{A[i - 1][c]}_{\text{Case 1}}, \underbrace{A[i - 1][c - s_i] + v_i}_{\text{Case 2}} \}
return A[n][C] // solution to largest subproblem$ 
```

Item	Value	Size
1	3	4
2	2	3
3	4	2
4	4	3

6	0	3	3	7	8
5	0	3	3	6	8
4	0	3	3	4	4
3	0	0	2	4	4
2	0	0	0	4	4
1	0	0	0	0	0
0	0	0	0	0	0
	0	1	2	3	4

Time complexity

Theorem 16.6 (Properties of Knapsack) *For every instance of the knapsack problem, the **Knapsack** algorithm returns the total value of an optimal solution and runs in $O(nC)$ time, where n is the number of items and C is the knapsack capacity.*

Reconstruction

Knapsack Reconstruction

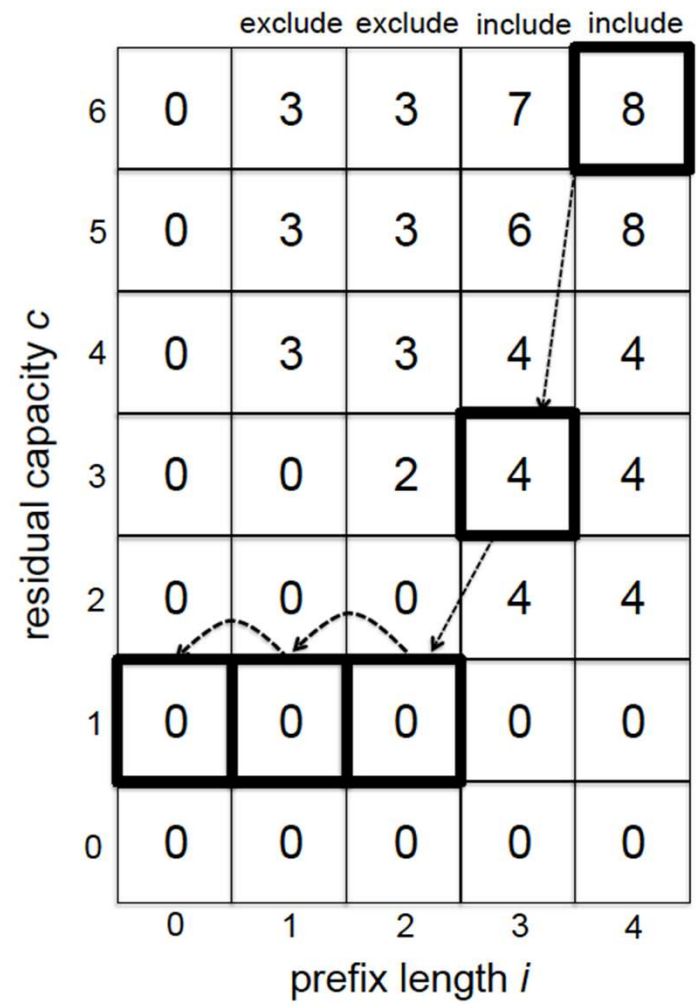
Input: the array A computed by the Knapsack algorithm with item values $v_1, v_2, \dots, v_n$, item sizes $s_1, s_2, \dots, s_n$, and knapsack capacity C .

Output: an optimal knapsack solution.

```
 $S := \emptyset$  // items in an optimal solution
 $c := C$  // remaining capacity
for  $i = n$  downto 1 do
    if  $s_i \leq c$  and  $A[i-1][c-s_i] + v_i \geq A[i-1][c]$  then
         $S := S \cup \{i\}$  // Case 2 wins, include  $i$ 
         $c := c - s_i$  // reserve space for it
    // else skip  $i$ , capacity stays the same
return  $S$ 
```

The Knapsack Reconstruction postprocessing step runs in $O(n)$ time (with $O(1)$ work per iteration of the main loop), which is much faster than the $O(nC)$ time used to fill in the array in the Knapsack

Item	Value	Size
1	3	4
2	2	3
3	4	2
4	4	3



s=4	s=3	s=2	s=3
\$3	\$2	\$4	\$4

Practice Problem

Problem 16.4 (*S*) Consider an instance of the knapsack problem with five items:

Item	Value	Size
1	1	1
2	2	3
3	3	2
4	4	5
5	5	4

and knapsack capacity $C = 9$. What are the final array entries of the Knapsack algorithm from Section 16.5, and which items belong to the optimal solution?

Solution 2nd, 3rd and 5th items

Capacity

9	0	1	3	6	8	10
8	0	1	3	6	8	9
7	0	1	3	6	7	9
6	0	1	3	6	6	8
5	0	1	3	5	5	6
4	0	1	3	4	4	5
3	0	1	2	4	4	4
2	0	1	1	3	3	3
1	0	1	1	1	1	1
0	0	0	0	0	0	0
	0	1	2	3	4	5

items

	\$1	\$2	\$3	\$4	\$5
size	1	3	2	5	4